Auto Ticket Classification using Flow Designer

A ServiceNow project for School IT Support

Item	Details
Platform	ServiceNow (Personal Developer Instance)
Tool used	Flow Designer
Custom table	Incident WorkFlow (u_incident_workflow)
Flow name	Auto Classify School IT Tickets
Update set	Project Update Set (Application: Global)

1. Introduction

Schools receive many IT complaints every day, such as Wi-Fi not working, projector problems, forgotten passwords and slow computers. When each ticket is sorted by hand, it takes time and mistakes happen. This project uses ServiceNow Flow Designer to classify every new ticket automatically.

When a student or staff member raises a ticket, the flow reads the short description, finds a keyword, and fills in the Category, Subcategory and Assigned Group without any manual work.

2. Objectives and Scope

Objectives

- Store all school IT tickets in one custom table with structured fields.
- Classify tickets automatically by Category and Subcategory.
- Route each ticket to the correct assignment group.
- Reduce response time and remove manual sorting.
- Keep all changes safely recorded in a Local Update Set.

How the solution works

The solution has four parts that work together:

Part	Purpose
Update Set	Records every configuration change so it can be moved or reviewed.
Custom Table	Stores the tickets with all required fields and choices.
Flow Designer flow	Runs when a ticket is created and classifies it.
Testing	Confirms each keyword sets the correct category and group.

Process flow

Ticket created > Flow triggers > Short description checked for keywords > Category, Subcategory and Group updated > Ticket ready for the support team.

3. Step 1: Create the Update Set

An update set captures every change made in the instance. Create it first so all later work (table, fields, flow) is recorded in it.

1. Log in to your ServiceNow instance.
2. Click **All** and search for **Update Sets**.
3. Under System Update Sets, click **Local Update Sets**.
4. Click the **New** button at the top right.
5. Enter the values shown in the table below.
6. Click **Submit** to save the record.
7. Open the new record and, under Related Links, click **Make This My Current Set**.
8. The created update set now appears as Current.

Field	Value
Name	Project Update Set
State	In progress
Application	Global

To confirm, click the gear icon, open the Developer tab, and check that Project Update Set is selected as the current set.

4. Step 2: Create the Custom Table

The custom table stores all ticket records that the flow will classify.

1. Click **All** and search for **Tables**.
2. Under System Definition, click **Tables**.
3. Click **New** at the top right of the list.
4. Label: **Incident WorkFlow**. The Name fills in automatically as u_incident_workflow.
5. Uncheck **Create module**.
6. Click **Submit** to save the form.
7. Open the table again and go to the **Controls** tab.
8. Tick the **Auto Number** checkbox to true.
9. Click **Update** to save the form again.

Tip: finding the table later

If the table does not show in the Tables list, search using the Label column filter, or open **u_incident_workflow.list** from the navigator. If the system says the table name is already in use, the table was created successfully and should not be created again.

5. Step 3: Create the Table Fields

Open the table and use the Columns section to click **New** for each field below.

Field Label	Type	Reference / Choices
Number	Auto Number	-
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	-
Description	String	-
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

Adding the choices

1. Create the Choice column and click Submit.
2. Reopen the column and click **New** in the Choices related list.
3. Enter the Label and Value (for example Network / network) and submit.
4. Repeat for every choice listed in the table above.

Subcategory dependency

Set the Subcategory column's **Dependent field** to Category, then give each choice a dependent value: Wi-Fi depends on Network, Projector on Hardware, Forgot Password on Access, and Slow Computer on Performance.

6. Step 4: Build the Flow and Trigger

Flow Designer lets you build automation without writing code. The flow starts when a new ticket is created.

1. Click **All**, search **Flow Designer**, and open **Flow Designer** under Process Automation.
2. Click **New** and choose **Flow**.
3. Flow name: **Auto Classify School IT Tickets**. Application: Global.
4. Click **Build flow**.
5. Click **Add a trigger** and choose **Created** under Record.
6. Set Table to **Incident WorkFlow [u_incident_workflow]**.
7. Leave the Condition empty so every new ticket runs the flow.
8. Click **Done** to save the trigger.

If Category is missing in the condition list

The Condition field only lists columns that exist on the trigger table. If Category is not shown, check that the column was added to Incident WorkFlow, then refresh the Flow Designer page and reselect the table.

7. Step 5: Classification Logic

Add an **If** block for each category. Each block checks whether the Short Description contains a keyword, then updates the ticket.

Keywords in Short Description	Category	Subcategory	Assigned Group
wifi, internet, network	Network	Wi-Fi	Network Support
projector, screen	Hardware	Projector	Hardware Support
password, login	Access	Forgot Password	Access Support
slow, hang, freeze	Performance	Slow Computer	Performance Support

Steps for each rule

1. Click the **+** icon under the trigger, choose **Flow Logic** and then **If**.
2. Drag the Short Description data pill from the trigger into the condition.
3. Choose the operator **contains** and type the keyword. Use **OR** to add more keywords.
4. Inside the If block, click **+** and choose the action **Update Record**.
5. Record: the trigger's Incident WorkFlow Record pill.
6. Click **Add field value** and set Category, Subcategory and Assigned Group.
7. Use **Else If** to add the next category and repeat.

Note: the group names above are examples. Use the groups that exist in your instance, or create them under User Administration > Groups.

8. Testing, Activation and Conclusion

Testing the flow

1. In Flow Designer, click **Test** at the top right.
2. Choose to run the flow on a new record and create a ticket, for example: Short Description = "Wi-Fi not working in Lab 2".
3. Click **Run Test** and open the execution details.
4. Check that Category = Network, Subcategory = Wi-Fi, and the group is assigned.
5. Repeat the test for the projector, password and slow computer keywords.
6. Click **Activate**. The status changes from Inactive to Active.

Common problems and fixes

Problem	Fix
Table not found in list	Search by Label, or open u_incident_workflow.list directly.
Table name already in use	The table exists. Do not submit again; open it instead.
Category missing in Flow condition	Add the column to the table, then refresh Flow Designer.
Flow did not run	Check the flow is Active and the trigger table is correct.

Conclusion

This project shows how Flow Designer can automate ticket handling without any coding. New school IT tickets are classified and routed within seconds, which improves response time and reduces manual effort. The same approach can be extended with priority rules, email notifications and SLA tracking.